Beaches_Bikes_Weather

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Importing Data and Libraries

```r
library(opendatatoronto)
```

```
Warning: package 'opendatatoronto' was built under R version 4.5.2
```

```r
library(tidyverse)
```

```
Warning: package 'purrr' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(sf)
```

```
Warning: package 'sf' was built under R version 4.5.2

Linking to GEOS 3.13.1, GDAL 3.11.4, PROJ 9.7.0; sf_use_s2() is TRUE
```

```r
library(leaflet)
```

```
Warning: package 'leaflet' was built under R version 4.5.2
```

```r
library(skimr)
```

```
Warning: package 'skimr' was built under R version 4.5.2
```

```r
library(janitor)
```

```
Warning: package 'janitor' was built under R version 4.5.2


Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(lubridate)
library(tidytext)
```

```
Warning: package 'tidytext' was built under R version 4.5.2
```

```r
library(dplyr)
library(broom)
```

```
Warning: package 'broom' was built under R version 4.5.2
```

```r
library(readr)
library(ggplot2)
```

```r
df <- read.csv("../01_Data/beaches_bikes.csv")
```

Bike/Weather Analysis

```r
Toronto_Daily_Temp <- read_csv("../01_Data/Toronto_Daily_Temp.csv")
```

```
Rows: 8627 Columns: 36
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (16): STATION_NAME, ID, PROVINCE_CODE, MEAN_TEMPERATURE_FLAG, MIN_TEMPE...
dbl  (16): x, y, CLIMATE_IDENTIFIER, LOCAL_YEAR, LOCAL_MONTH, LOCAL_DAY, MEA...
lgl   (3): TOTAL_SNOW, DIRECTION_MAX_GUST, SPEED_MAX_GUST
dttm  (1): LOCAL_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

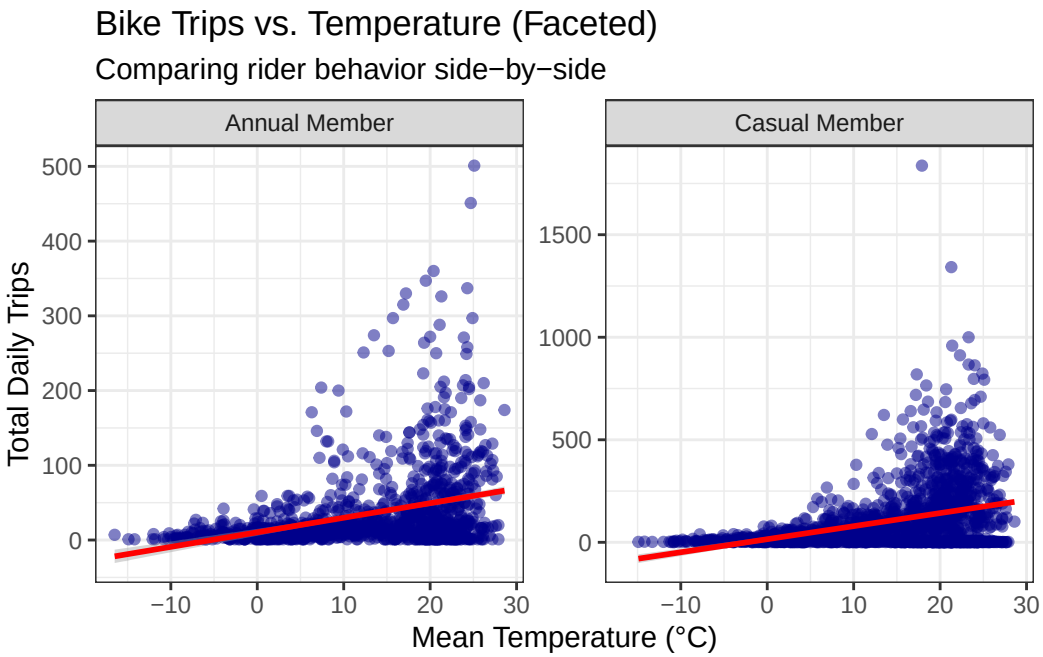
```r
Toronto_Daily_Temp <- clean_names(Toronto_Daily_Temp)
Toronto_Daily_Temp <- Toronto_Daily_Temp |> filter(local_year > 2020)

# parse date so it is compatible with the bike share data
temp_clean <- Toronto_Daily_Temp |>
  mutate(

    date = as_date(local_date),

    month_name = month(date, label = TRUE)
  ) |>
  # sort by the new datetime column
  arrange(date) |>

  filter(!is.na(date))
```

```r
# join temperature data to bike share
combined_data <- df |>
  mutate(join_date = as_date(end_date_actual)) |>
  inner_join(temp_clean, by = c("join_date" = "date")) |>
  mutate(year = year(join_date))
```

```r
# Prepare the daily data
daily_data_by_type <- combined_data %>%
  group_by(start_date_actual, mean_temperature, User.Type) %>%
  summarise(
    total_trips = n(),
```

```r
    .groups = "drop"
  )

# plot with Facets
ggplot(daily_data_by_type, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +

  facet_wrap(~ User.Type, scales = "free_y") +

  labs(
    title = "Bike Trips vs. Temperature (Faceted)",
    subtitle = "Comparing rider behavior side-by-side",
    x = "Mean Temperature (°C)",
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

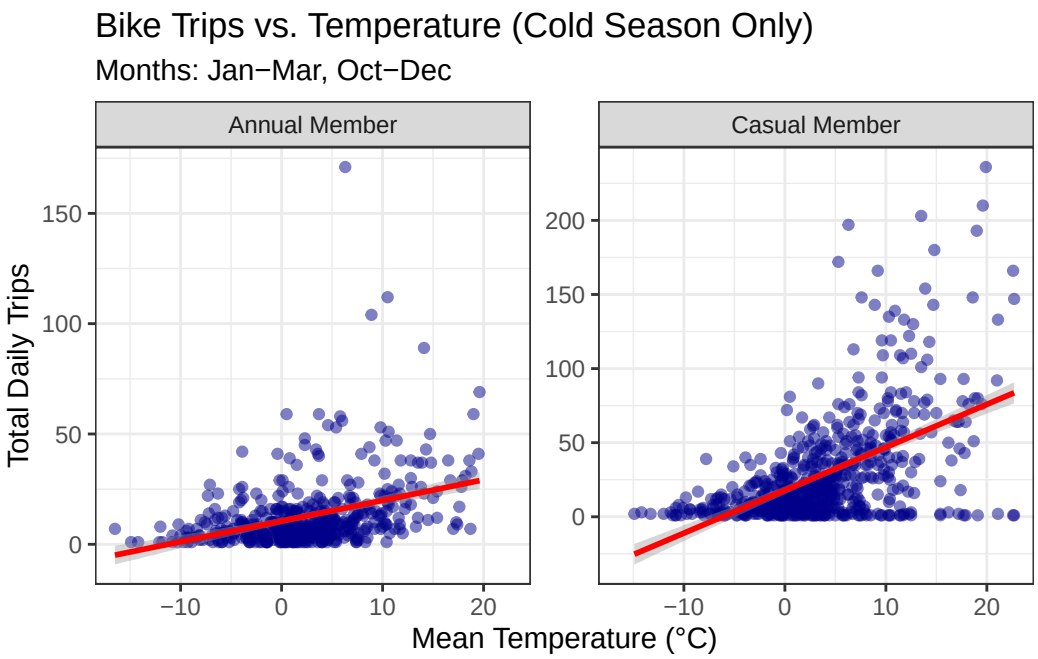
```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 18 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 18 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
# Prepare the daily data
filtered_daily_data <- combined_data %>%
  # Aggregate to daily counts per user type
  group_by(start_date_actual, mean_temperature, User.Type) %>%
  summarise(total_trips = n(), .groups = "drop") %>%

  # filter for shoulder season
  filter(month(start_date_actual) %in% c(1, 2, 3, 10, 11, 12))

# plot
ggplot(filtered_daily_data, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ User.Type, scales = "free_y") +
  labs(
    title = "Bike Trips vs. Temperature (Cold Season Only)",
    subtitle = "Months: Jan-Mar, Oct-Dec",
    x = "Mean Temperature (°C)",
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 7 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
# run correlation test for each group
correlation_results <- filtered_daily_data %>%
  group_by(User.Type) %>%
  summarise(

    correlation = cor(mean_temperature, total_trips, use = "complete.obs"),

    p_value = cor.test(mean_temperature, total_trips)$p.value,

    n_obs = n()
  )

# show results
print(correlation_results)
```

```
# A tibble: 2 x 4
```

```
  User.Type     correlation  p_value n_obs
  <chr>               <dbl>    <dbl> <int>
1 Annual Member       0.381 3.97e-17   458
2 Casual Member       0.543 3.42e-49   630
```

Using 3 Clusters

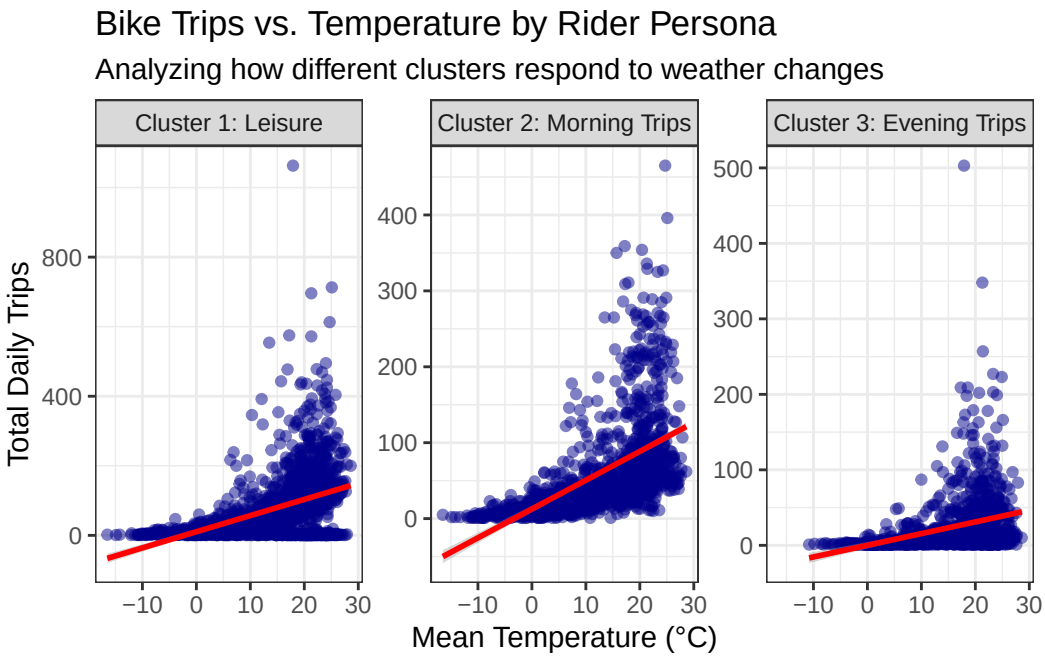
Compare 3 clusters rather than user type

```r
set.seed(2453)
# clean data
clustering_temp <- combined_data %>%
  filter(!is.na(Trip..Duration) & !is.na(start_hour_of_day)) %>%
  filter(Trip..Duration < 7200)

kmeans_scaled <- scale(clustering_temp %>% select(Trip..Duration, start_hour_of_day))
trip_clusters <- kmeans(kmeans_scaled, centers = 3, nstart = 25)
clustering_temp$Cluster <- as.factor(trip_clusters$cluster)

# prepare dily data
daily_data_cluster <- clustering_temp %>%
  # Group by date, temperature, and Cluster
  group_by(start_date_actual, mean_temperature, Cluster) %>%
  summarise(total_trips = n(), .groups = "drop")


# Summarise the profiles including User Type Proportions
cluster_profiles <- clustering_temp |>
  group_by(Cluster) |>
  summarise(
    Total_Trips = n(),
    Avg_Start_Hour = round(mean(start_hour_of_day), 1),
    Avg_Duration_Mins = round(mean(Trip..Duration) / 60, 1),

    # Calculate the percentage of each member type
    Pct_Annual = round(sum(User.Type == "Annual Member") / n() * 100, 1),
    Pct_Casual = round(sum(User.Type == "Casual Member") / n() * 100, 1)
  ) |>
  arrange(Cluster)

# Print the resulting table
print(cluster_profiles)
```

```
# A tibble: 3 x 6
  Cluster Total_Trips Avg_Start_Hour Avg_Duration_Mins Pct_Annual Pct_Casual
  <fct>         <int>          <dbl>             <dbl>      <dbl>      <dbl>
1 1            112980           17.9              21.1       20.7       79.3
2 2             73358           10.9              20.5       22.9       77.1
3 3             27009           15.6              61.5        3.5       96.5
```

```
# plot with facets by cluster
ggplot(daily_data_cluster, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +

  facet_wrap(~ Cluster, scales = "free_y",
             labeller = as_labeller(c("1" = "Cluster 1: Leisure",
                                      "2" = "Cluster 2: Morning Trips",
                                      "3" = "Cluster 3: Evening Trips"))) +

  labs(
    title = "Bike Trips vs. Temperature by Rider Persona",
    subtitle = "Analyzing how different clusters respond to weather changes",
    x = "Mean Temperature (°C)",
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 26 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 26 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
set.seed(2453)
# clean data
clustering_temp <- combined_data %>%
  filter(!is.na(Trip..Duration) & !is.na(start_hour_of_day)) %>%
  filter(Trip..Duration < 7200)

kmeans_scaled <- scale(clustering_temp %>% select(Trip..Duration, start_hour_of_day))
trip_clusters <- kmeans(kmeans_scaled, centers = 3, nstart = 25)
clustering_temp$Cluster <- as.factor(trip_clusters$cluster)

# prepare data for shoulder season
filtered_daily_data_cluster <- clustering_temp %>%
  # Filter for shoulder season
  filter(month(start_date_actual) %in% c(1, 2, 3, 10, 11, 12)) %>%
  # Group by date, temperature, and Cluster
  group_by(start_date_actual, mean_temperature, Cluster) %>%
  summarise(total_trips = n(), .groups = "drop")


# Summarise the profiles including User Type Proportions
cluster_profiles <- clustering_temp |>
  filter(month(start_date_actual) %in% c(1, 2, 3, 10, 11, 12)) %>%
  group_by(Cluster) |>
```

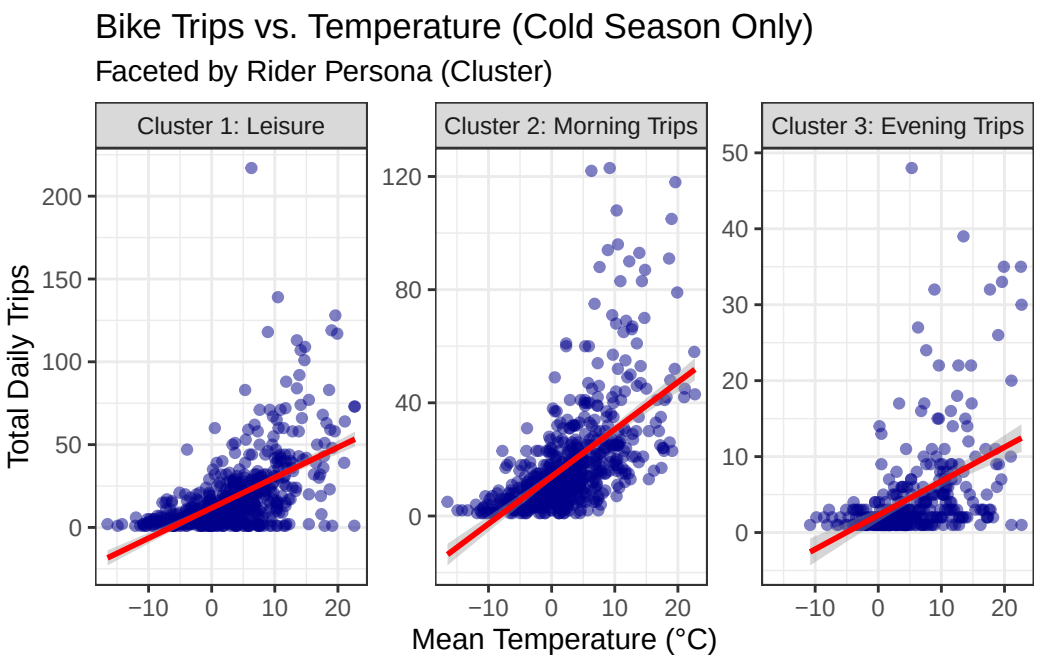
```r
  summarise(
    Total_Trips = n(),
    Avg_Start_Hour = round(mean(start_hour_of_day), 1),
    Avg_Duration_Mins = round(mean(Trip..Duration) / 60, 1),

    # Calculate the percentage of each member type
    Pct_Annual = round(sum(User.Type == "Annual Member") / n() * 100, 1),
    Pct_Casual = round(sum(User.Type == "Casual Member") / n() * 100, 1)
  ) |>
  arrange(Cluster)

# Print the resulting table
print(cluster_profiles)
```

```
# A tibble: 3 x 6
  Cluster Total_Trips Avg_Start_Hour Avg_Duration_Mins Pct_Annual Pct_Casual
  <fct>         <int>          <dbl>             <dbl>      <dbl>      <dbl>
1 1             11038           17.4              18.6       26.6       73.4
2 2             11106           10.7              18.9       25.9       74.1
3 3              1614           14.8              59.3        6.3       93.7
```

```r
# plot result
ggplot(filtered_daily_data_cluster, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ Cluster, scales = "free_y",
             labeller = as_labeller(c("1" = "Cluster 1: Leisure",
                                      "2" = "Cluster 2: Morning Trips",
                                      "3" = "Cluster 3: Evening Trips"))) +
  labs(
    title = "Bike Trips vs. Temperature (Cold Season Only)",
    subtitle = "Faceted by Rider Persona (Cluster)",
    x = "Mean Temperature (°C)",
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 11 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 11 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
# run the correlation test for each cluster group
correlation_results_cluster <- filtered_daily_data_cluster %>%
  group_by(Cluster) %>%
  summarise(
    correlation = cor(mean_temperature, total_trips, use = "complete.obs"),

    p_value = cor.test(mean_temperature, total_trips)$p.value,

    n_days = n()
  )

# show results
print(correlation_results_cluster)
```

```
# A tibble: 3 x 4
  Cluster correlation  p_value n_days
  <fct>         <dbl>    <dbl>  <int>
1 1             0.549 2.98e-50    627
2 2             0.586 2.44e-55    590
3 3             0.437 7.44e-17    334
```

Only using Leisure Cluster Data

```
df2 <- read_csv("../01_Data/leisure_dataset.csv")
```

```
Rows: 27009 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr   (8): Start.Time, Start.Station.Name, End.Time, End.Station.Name, User....
dbl  (10): Trip.Id, Trip..Duration, Start.Station.Id, End.Station.Id, Bike.I...
dttm  (2): datetime_start, datetime_end
date  (2): start_date_actual, end_date_actual

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# join temperature data to bike share leisure cluster
combined_data <- df2 |>
  mutate(join_date = as_date(end_date_actual)) |>
  inner_join(temp_clean, by = c("join_date" = "date")) |>
  mutate(year = year(join_date))
```

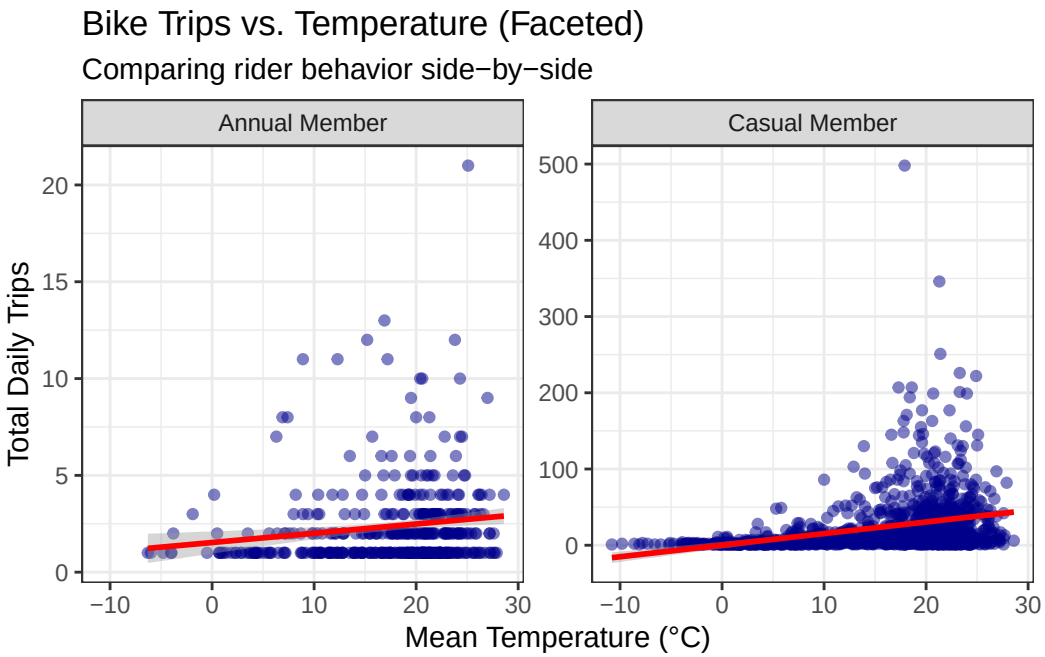
```
# Prepare the daily data
daily_data_by_type <- combined_data %>%
  group_by(start_date_actual, mean_temperature, User.Type) %>%
  summarise(
    total_trips = n(),
    .groups = "drop"
  )

# plot with facets
ggplot(daily_data_by_type, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +

  facet_wrap(~ User.Type, scales = "free_y") +

  labs(
    title = "Bike Trips vs. Temperature (Faceted)",
    subtitle = "Comparing rider behavior side-by-side",
    x = "Mean Temperature (°C)",
```

```
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 10 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 10 rows containing missing values or values outside the scale range
(`geom_point()`).
```



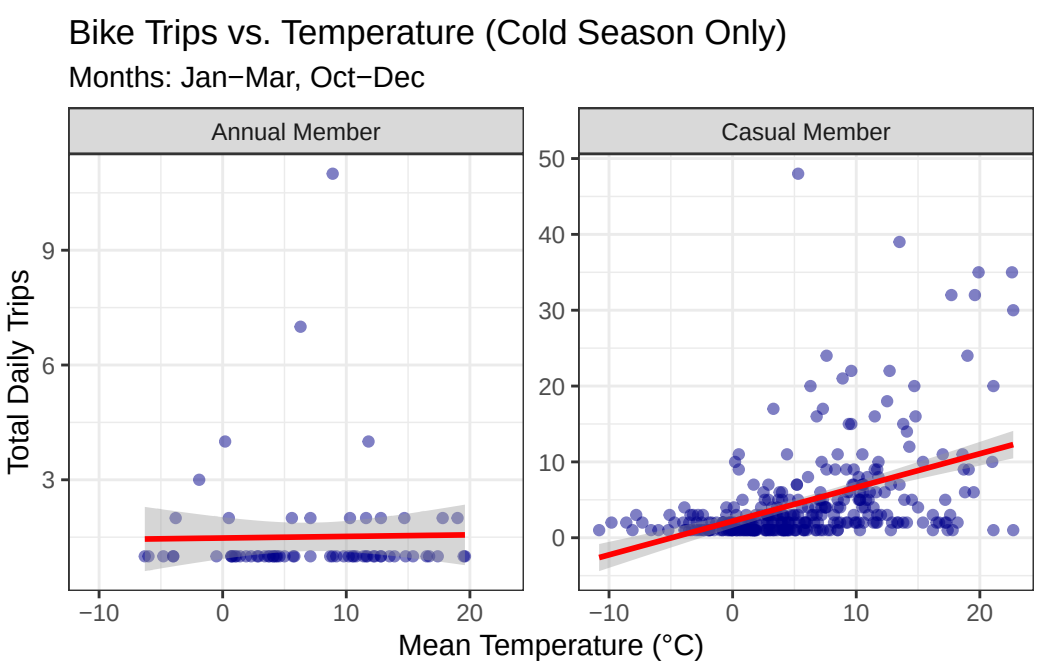

```
# Prepare the daily data
filtered_daily_data <- combined_data %>%
  # Aggregate to daily counts per user type
  group_by(start_date_actual, mean_temperature, User.Type) %>%
  summarise(total_trips = n(), .groups = "drop") %>%

  #filter for shoulder season
  filter(month(start_date_actual) %in% c(1, 2, 3, 10, 11, 12))
```

```r
# plot results
ggplot(filtered_daily_data, aes(x = mean_temperature, y = total_trips)) +
  geom_point(alpha = 0.5, color = "darkblue") +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ User.Type, scales = "free_y") +
  labs(
    title = "Bike Trips vs. Temperature (Cold Season Only)",
    subtitle = "Months: Jan-Mar, Oct-Dec",
    x = "Mean Temperature (°C)",
    y = "Total Daily Trips"
  ) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 4 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 4 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
# Run the correlation test for each group
correlation_results <- filtered_daily_data %>%
  group_by(User.Type) %>%
  summarise(
    correlation = cor(mean_temperature, total_trips, use = "complete.obs"),

    p_value = cor.test(mean_temperature, total_trips)$p.value,

    n_obs = n()
  )

# show results
print(correlation_results)
```

```
# A tibble: 2 x 4
  User.Type     correlation  p_value n_obs
  <chr>               <dbl>    <dbl> <int>
1 Annual Member      0.0183 8.83e- 1    68
2 Casual Member      0.443  1.56e-16   317
```